DO WE REALLY KNOW AUCKLANDS EARTHQUAKE RISK?
A CASE STUDY ON THE MANGATANGI FAULT

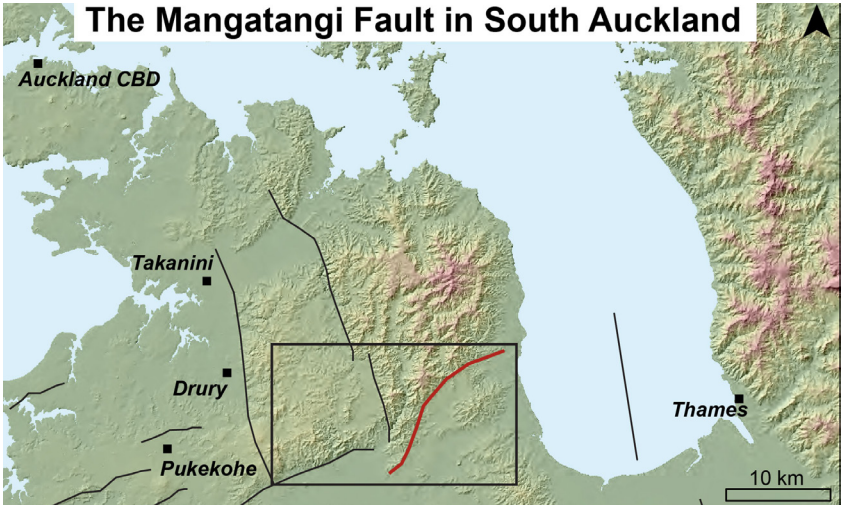
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Auckland is home to almost a third of Aotearoa New Zealand's population and it drives much of the country's economy. We know its volcanoes well. Fifty-odd cones scattered across the isthmus, Rangitoto sitting in the harbor like a postcard. But its faults are a different story. For a city this size we know remarkably little about what's lurking beneath it.

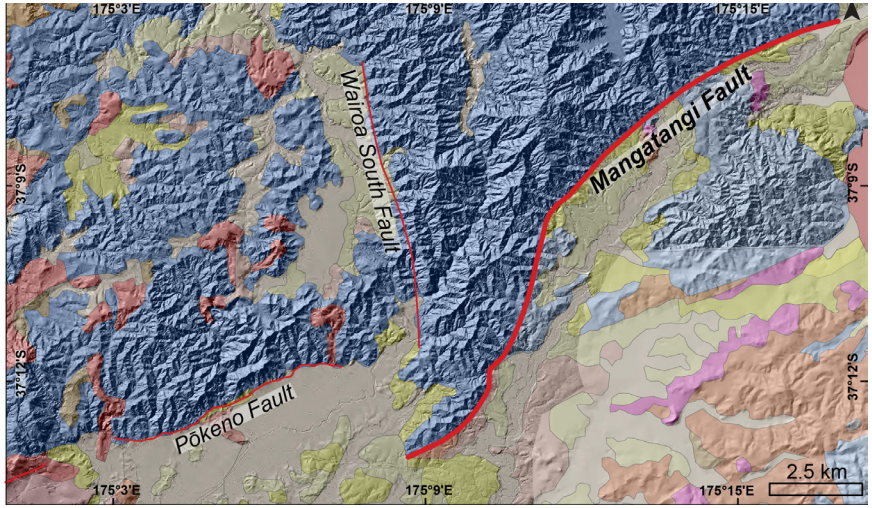
Part of the problem is that many faults in Aotearoa New Zealand move so infrequently that "active" doesn't mean what people assume. A fault here is classified as active if it has ruptured at some point in the last 125,000 years (e.g. Stirling et al., 2012). With gaps of tens of thousands of years between ruptures, there's plenty of time for erosion and deposition to quietly erase the evidence. And in a place like Auckland, where the landscape is so young and has been continually reshaped by volcanism, that evidence can be even harder to find. The very thing that makes Auckland's geology so distinctive (its volcanic field) may also be quietly burying or blurring the faults underneath it. Even the seemingly generous 125,000 year classification can have its limitations, in that some faults can produce major earthquakes on even longer time scales in these low seismicity regions.

Recent LiDAR mapping by Bland et al. (2023) revealed several fault scarps no one had mapped before and most of which were subsequently added to the New Zealand Active Faults Database. This was a huge step forward, giving us a far better picture of Auckland's seismic prehistory than we had previously. But fault mapping and paleoseismic investigations are two very different things. None of these newly identified faults had a dated rupture. Their "active" classification was derived from how fresh they looked in the landscape, a common and reasonable proxy for activity, but a proxy nonetheless. The geomorphology told us where to look, but it couldn't tell us when these faults last moved. Or how often.

Our study (Martin et al., 2026), just published in the New Zealand Journal of Geology and Geophysics, provides the first radiometrically



Modified from Martin et al. (2026)



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dated fault movement anywhere near Auckland. We investigated the Mangatangi Fault, which runs along the southwestern edge of the Hunua Ranges about 50 km south of Auckland city. Radiocarbon dating shows that it has ruptured within the last 10,000 years, well within the 125,000-year "active" threshold. For a fault that extends just over 18 km with a well-preserved scarp a couple of metres high, the Mangatangi Fault could be capable of a maximum magnitude 6.8 earthquake (or more), with a recurrence interval somewhere between 8,000 and 127,000 years. And while the Mangatangi Fault has had all the attention, it's neighbours like the Pōkeno Fault show similar geomorphology in the landscape and could share a similar history, preserving comparable scarps under the same erosional and depositional conditions.

So what would an earthquake in this part of Auckland actually look like if it happened? Modelling by Dempsey et al. (2021) on nearby faults gives us a sense: severe shaking near the source, with the Hunua dams (which supply much of Auckland's water) sitting right in the firing line, and strong shaking reaching into South Auckland and possibly further into the central city depending on the ground conditions. The closest thing we have to a real-world comparison is the 1891 Port Waikato earthquake, a magnitude ~ 6.0 event that broke windows, toppled chimneys, and reportedly knocked schoolchildren off their chairs. A rupture along the Mangatangi Fault could be larger than that, and it would be hitting a very different Auckland. One with hundreds of thousands more people, and a lot more roads, rail lines, and critical infrastructure sitting in harm's way.

Late last year, the government confirmed that Auckland, Northland, and the Chatham Islands will be removed entirely from the earthquake-prone building regime on the grounds that these regions pose a low seismic risk. Auckland's mayor has been a vocal supporter of this action, pointing to a long stretch of time since Auckland last had a significant earthquake. We understand the reasoning. The financial burden of the current system on building owners is real and targeting resources at genuine risk makes sense in principle. But our findings sit a little awkwardly next to that decision. We've just shown that a fault south of the city has moved in the Holocene and that we know far less about Auckland's seismic hazard than these kinds of policy calls assume.

It's hard to talk about "low seismic hazard" regions in Aotearoa New Zealand without thinking of Ōtautahi Christchurch. Before 2010, the Greendale Fault wasn't on any map (Quigley et al., 2010a; 201b). It was buried under thousands of years of river gravel with no surface trace. That earthquake, and the devastating one that followed in February 2011 weren't the result of bad science, they were the result of incomplete information on faults that the landscape itself had hidden. We know every "low hazard" region in New Zealand gets compared to Ōtautahi. But that's kind of the point. It became the obvious comparison precisely because it was so unexpected and so devastating. If Auckland's faults are being similarly obscured (by erosion, sediment, or the young and active volcanic landscape), then "low hazard" might simply mean "low information", until the day it doesn't. The Mangatangi Fault is capable of producing a larger earthquake than the magnitude 6.3 in L'Aquila, Italy in 2009 which killed over 300 people (Walters et al., 2009; Alexander, 2013). In contrast, the L'Aquila region is classified as medium-high seismic hazard (Stucchi et al., 2011), while Auckland is classified as low. That's not necessarily wrong. New Zealand lives with so many hazards that our risk tolerance is calibrated differently. "Low hazard" here means something different to "low hazard" in a lot of other countries.

So how confident are we really in Auckland's seismic hazard picture? Honestly not very. This study is just one fault dated for the first time. There are other newly mapped faults nearby with no reported slip rates, no reported recurrence intervals, and we have no idea when they last moved. The Mangatangi Fault may sit quietly for tens of thousands of years. But it's a reminder that we've been treating an absence of data as if it were evidence of safety, and those aren't the same thing. Auckland keeps expanding south closer to these potentially active (and in the case of the Mangatangi, confirmed active) faults, while our understanding of them stays the same as it was two decades ago. Before we lock in decisions that assume Auckland is low risk for good, we owe it to the city to go and find out properly. We can't keep planning around faults we haven't dated.

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